# The Cognitive Architecture of Morality: Kohlberg's Framework and its Application in Artificial Intelligence Benchmarking

## Introduction

As artificial intelligence transitions from deterministic, rule-based computation to generative, agentic systems capable of autonomous decision-making, the necessity for robust ethical alignment has become a paramount concern in computer science and cognitive psychology. Large language models are increasingly deployed in high-stakes environments—ranging from healthcare diagnostics and educational tutoring to legal adjudication and content moderation—where their outputs can precipitate profound societal consequences. To evaluate whether these highly capable systems are safe, reliable, and aligned with human values, researchers have increasingly turned to established paradigms in developmental psychology. Chief among these frameworks is Lawrence Kohlberg’s theory of moral development.

For decades, Kohlberg's cognitive-developmental framework has served as the gold standard for assessing how human beings mature from obedience-driven infants into principled, autonomous moral agents. Today, this exact developmental map is being repurposed to measure the artificial moral compass of machine learning models. By subjecting large language models to psychometric evaluations like the Defining Issues Test and specialized ethical benchmarks, the computational linguistics community is attempting to quantify the moral reasoning capabilities of artificial intelligence.

This analysis provides an exhaustive examination of Kohlberg’s moral psychology theory, tracing its historical origins, theoretical architecture, and the pivotal feminist and psychometric critiques that have shaped current moral psychology. Furthermore, the evaluation investigates the intersection of this cognitive framework with artificial intelligence, exploring how Kohlberg's stages are utilized to benchmark large language models, the phenomenon of moral ventriloquism in machine outputs, and the mathematical alignment techniques—such as Direct Preference Optimization and Constitutional AI—that engineer these ethical behaviors. Ultimately, this examination assesses the promises and profound paradoxes inherent in benchmarking the morality of artificial systems.

## The Origins and Evolution of Kohlberg's Theory

The foundation of modern moral psychology was initially laid by the Swiss psychologist Jean Piaget, who proposed that moral development in children occurs in distinct stages inextricably linked to cognitive maturation, life experiences, and active reasoning. Building directly upon Piaget’s cognitive-developmental approach, American psychologist Lawrence Kohlberg introduced his own expanded, more rigorous model in his 1958 doctoral thesis.

Kohlberg’s fundamental premise was that morality is inherently cognitive in nature. His theory focuses not on the final moral choice an individual makes, but on the underlying reasoning

process and structural logic that leads to that choice. To study this phenomenon, Kohlberg developed the Moral Judgment Interview, a methodology that involved presenting subjects—ranging from young children to adolescents and adults—with complex, hypothetical moral dilemmas where competing ethical principles were fundamentally at odds.

The most famous of Kohlberg's psychometric instruments is the Heinz Dilemma. The scenario describes a man named Heinz living in Europe, whose wife is dying from a rare form of cancer. A local pharmacist has recently discovered a radium-based drug that could cure her, but the pharmacist is charging $2,000 for a single dose that cost him only $200 to manufacture. Heinz cannot afford the drug, even after borrowing money from acquaintances. He begs the pharmacist to sell the medication cheaper or allow him to pay later, but the pharmacist categorically refuses, citing his right to profit from his own invention. Desperate and running out of time, Heinz breaks into the pharmacy after hours and steals the drug to save his wife.

Following the presentation of the narrative, subjects are asked whether Heinz was right or wrong to steal the drug, and, crucially, why they hold that belief. Kohlberg also utilized follow-up questions to probe the boundaries of the subject's moral logic, asking whether it would change anything if Heinz did not love his wife, what if the person dying was a complete stranger, and whether the police should arrest the chemist for murder if the woman ultimately died.

Kohlberg recognized early in his research that the binary choice—to steal or not to steal—was irrelevant to developmental measurement. The true indicator of moral maturity lay entirely in the justification provided. A subject might advocate stealing because Heinz would face blame from his family if his wife died, which represents a low-level, consequence-based reason. Conversely, another subject might advocate stealing because the inherent, universal value of a human life supersedes the societal law of property rights, which represents a high-level, principled reason. By meticulously mapping these justifications across longitudinal studies, Kohlberg identified a universal, sequential progression of moral logic.

# The Architecture of Moral Development

Kohlberg conceptualized moral development as a hierarchical progression through three primary levels, each containing two distinct stages, resulting in a six-stage framework. The theory dictates that individuals must progress through these stages sequentially; cognitive constraints make it impossible to skip a stage, and comprehension of higher stages is cumulative, integrated, and increasingly complex.

## Level 1: Pre-Conventional Morality

At the pre-conventional level, which is typically observed in children under the age of nine but can persist into adulthood, morality is entirely externally controlled. Individuals at this stage do not yet internalize societal conventions of right and wrong. Instead, they view morality as something external to themselves, judging the ethics of an action strictly by its direct physical consequences in an egocentric manner.

Stage 1 is characterized as the Obedience and Punishment Orientation. Behavior is dictated almost exclusively by a desire to avoid punishment. The individual exhibits blind compliance to powerful authority figures, believing that rules are fixed, absolute, and handed down by those in power. The well-being of other people is not acknowledged at this stage. In the context of the Heinz dilemma, a Stage 1 respondent might argue that Heinz should not steal because he will be caught and sent to prison. Conversely, they might argue that he should steal the drug, but

must be severely punished because stealing is inherently a rule violation. The underlying logic remains entirely egocentric, equating the severity of a punishment with the inherent badness of the act itself.
Stage 2 represents the Instrumental Purpose or Self-Interest and Exchange Orientation. At this stage, cognitive growth allows individuals to recognize that there is not just one right view handed down by authorities; they realize that different people possess different perspectives and needs. Action is judged by how well it satisfies personal needs or results in a direct reward. Relationships are viewed in highly transactional terms, characterized by a quid pro quo or "You scratch my back, and I'll scratch yours" mentality. A Stage 2 justification for the Heinz dilemma might argue that Heinz should steal the drug because he needs his wife to take care of him, or, alternatively, that he should not steal it because he might spend more time in jail than he would ultimately get to spend with his surviving wife. Concern for others exists only to the extent that it furthers the individual's own interests.

## Level 2: Conventional Morality

During adolescence and early adulthood, individuals typically transition into the conventional level, where conformity to social rules, norms, and expectations becomes the paramount driver of moral reasoning. The psychological emphasis shifts away from pure self-interest toward the maintenance of interpersonal relationships and the preservation of broader societal order. Morality is internalized, but it remains heavily dependent on the approval of external socializing agents such as family, peers, and civic institutions.
Stage 3 is defined as the Good Interpersonal Relationships or Good Boy/Nice Girl Orientation. Moral reasoning at this stage is driven by a profound desire to be perceived as a good person and to live up to the expectations of one's immediate community. Actions are evaluated based on internal motives and interpersonal feelings such as love, empathy, and loyalty, rather than strict transactional exchange. A Stage 3 respondent typically supports Heinz's decision to steal, arguing that any loving husband would do whatever it takes to save his family. They frequently emphasize the pharmacist's negative traits, describing him as selfish and greedy, thereby justifying Heinz's theft as the action of a fundamentally good person driven by noble intentions.
Stage 4 transitions into the Law and Order Orientation. The individual's cognitive perspective broadens significantly from local, interpersonal relationships to society as a comprehensive whole. Morality is strictly defined by obeying societal laws, respecting institutional authority, and performing one's civic duties to avoid a breakdown of the social fabric. A Stage 4 thinker might express deep sympathy for Heinz's motives and tragic circumstances, but ultimately conclude that stealing is morally wrong. Their reasoning centers on the premise that the law applies to everyone equally, and if individuals were permitted to break the law whenever they felt they had a compelling personal reason, social order would disintegrate and society would descend into chaos.

## Level 3: Post-Conventional and Principled Morality

At the highest level of moral functioning—which Kohlberg empirically observed many adults never fully reach—individuals judge the rightness of an action based on abstract, self-chosen ethical principles. Rules and laws are no longer viewed as absolute mandates to be obeyed blindly; rather, they are seen as useful but flexible mechanisms that must actively serve human rights, universal justice, and the greater good.
Stage 5 is characterized as the Social Contract and Individual Rights Orientation. Individuals at

this stage view laws as social contracts designed to protect the greatest good for the greatest number of people. However, they recognize that when laws fail to protect individual fundamental rights or the overarching interests of the majority, those laws are deemed unjust and should be subject to democratic change or principled violation. A Stage 5 justification might assert that while the law against stealing is generally valid and necessary for societal trust, the fundamental human right to life unconditionally transcends property rights, making Heinz's theft a morally justified exception that highlights a fundamentally unfair economic and legal system.

Stage 6 represents the pinnacle of Kohlberg's framework: the Universal Ethical Principles Orientation. Moral decisions at this stage are driven entirely by internal, self-chosen principles of conscience that are universally applicable—such as the inherent dignity of human life, radical equality, and absolute justice. A Stage 6 thinker recognizes that civic rules are subordinate to universal human rights; a law that violates human dignity must necessarily be broken, regardless of the personal consequences or societal disruption. An example of Stage 6 reasoning is a whistleblower exposing corporate corruption because the universal principle of justice outweighs their personal risk of prosecution or termination. Kohlberg himself noted that he found it extremely challenging to identify participants who consistently displayed moral reasoning at this sixth stage, as it requires a high degree of abstract, philosophical cognition.

# Critiques and Paradigm Shifts: The Ethics of Care

While Kohlberg's theory was revolutionary, effectively establishing the cognitive-developmental approach to socialization and ethical reasoning, it faced severe methodological and theoretical criticism by the early 1980s. The most consequential challenge to Kohlberg's hegemony came from Carol Gilligan, a developmental psychologist and former research assistant to Kohlberg himself. Gilligan published the seminal work *In a Different Voice: Psychological Theory and Women's Development* in 1982, a text that Harvard University Press later described as the little book that started a revolution in gender studies and moral philosophy.

## The Gender Bias and the Measurement Critique

Kohlberg's original empirical data was derived from a twenty-year longitudinal study consisting exclusively of white, upper-class male subjects from the Chicago area. When his complex scoring manual was subsequently applied to broader, more diverse populations, researchers observed a persistent and troubling anomaly: female participants, on average, scored lower than their male counterparts. Women frequently plateaued at Stage 3, heavily emphasizing interpersonal relationships and care, while men routinely advanced to Stage 4 and Stage 5, emphasizing laws, rights, and abstract social contracts.

Kohlberg and other researchers initially interpreted this data to mean that women were less morally developed than men. Gilligan vehemently argued that this developmental disparity was not indicative of stunted moral growth in women, but rather a fundamental, architectural flaw in the measurement instrument itself. Kohlberg's entire framework was built upon an "Ethics of Justice"—a paradigm that conceptualizes ultimate moral maturity as the ability to operate autonomously, rank competing rights, rely on the legal system, and apply abstract, impartial rules to settle conflicts.

Gilligan posited that women frequently utilize a different, but equally mature, moral logic, which she termed the "Ethics of Care." This alternative framework views moral dilemmas not as mathematical problems involving competing individual rights, but as ruptures in an

interconnected web of relationships that require a response based on empathy, compassion, responsibility, and the strict prevention of harm. When confronted with the Heinz dilemma, male subjects typically treated it as a logic puzzle, deducing that the right to life legally outranks the right to property, and therefore Heinz must steal the drug. Female subjects, conversely, traced the relational network, noting that if Heinz goes to prison for theft, his sick wife will be left entirely uncared for, suggesting that further communication, compromise, and community intervention with the pharmacist is the only sustainable moral path. Because Kohlberg's manual prioritized abstract logic over relational sustainability, it systematically scored the justice-oriented response a full stage higher; Gilligan argued that the care-oriented subjects were answering a different, but equally sophisticated, moral question.

## Stages of the Ethics of Care

Alongside her critique of Kohlberg, Gilligan proposed a parallel developmental sequence within the care framework, outlining how relational moral reasoning matures over time through three distinct stages.

The first stage is Pre-Conventional Care, or the orientation toward individual survival. In this initial phase, care is directed solely at the self, focusing on individual survival in what is perceived as a hostile, isolating world. This corresponds to the egocentrism of Kohlberg's Stage 1, where the individual feels unable to look past their own immediate self-interest and survival needs.

The second stage is Conventional Care, defined as goodness as self-sacrifice. As the individual matures, they begin to equate moral goodness directly with selflessness. Care is directed toward everyone except the self. This stage aligns closely with societal expectations of traditional femininity, where asserting one's own needs is culturally viewed as selfish, and the individual actively seeks solutions where no one else is hurt, even if it requires significant personal detriment.

The third and final stage is Post-Conventional Care, representing the mature ethics of care. At this pinnacle, the individual realizes that sustainable caregiving inherently requires self-care. The moral web expands to comprehensively include both the self and others, governed by a universal principle of non-violence and harm reduction. The individual takes full responsibility for the consequences of their choices, sets healthy relational limits, and recognizes the deep interconnectedness of all parties involved.

## The Current Thinking and Modern Synthesis

Subsequent decades of psychological research have refined and slightly corrected Gilligan's original critique. The modern scientific consensus suggests that the Ethics of Care and the Ethics of Justice are not strictly or essentially gender-bound. Both sexes utilize both orientations depending heavily on the specific context of the dilemma and cultural variables. Gilligan's enduring contribution was demonstrating that Kohlberg's measurement stick was incomplete. True moral maturity is now understood as a dual-process integration of both frameworks: the capacity to hold fast to abstract principles of justice while remaining acutely attentive to contextual relationships and the practice of genuine care.

This evolution in moral psychology—from a rigid, justice-only hierarchy to a pluralistic understanding of moral reasoning—forms the critical theoretical backdrop for how modern computer scientists are now attempting to evaluate the ethical reasoning of artificial intelligence.

# The Intersection of Kohlberg's Theory and Artificial Intelligence

As artificial intelligence systems transition from passive, conversational chatbots to autonomous agents capable of making healthcare recommendations, moderating global speech, and executing financial logic, the field of AI alignment has encountered a critical imperative: how to imbue machines with human-aligned moral reasoning.

Researchers have increasingly realized that strict behavioral compliance is entirely insufficient for advanced AI. If an AI model is merely programmed with binary, hardcoded rules to avoid specific restricted behaviors, it acts much like a human child at Kohlberg's Stage 1—avoiding restricted outputs merely to prevent negative loss functions or algorithmic punishment, without internalizing the fundamental reason for the rule. This creates brittle computational systems that are highly susceptible to adversarial jailbreaks and context manipulation. To create robust, resilient artificial intelligence, developers are attempting to push large language models into the computational equivalent of post-conventional moral reasoning.

## Cognitive Morality Classification for AI Systems

Theoretical frameworks have recently been proposed to classify the cognitive morality of AI systems by using Kohlberg's stages as an analogical map for computational architectures. This involves translating human psychological stages into the structural capabilities of neural networks.

Level 1 represents Pattern Identification, which is functionally analogous to Kohlberg's pre-conventional level. Current statistical learning models, utilizing System 1 thinking, detect regularities and patterns in vast datasets. They can accurately classify text as toxic or safe based purely on statistical correlation, but they fundamentally lack explicit rule comprehension or moral abstraction.

Level 2 represents Rule Abstraction, functionally similar to Kohlberg's conventional stages of moral development. At this level, AI systems can infer explicit rules from provided examples and apply them modularly to new situations. They can follow explicit policy instructions, akin to Stage 4 Law and Order reasoning, operating as highly efficient rule-following algorithms. However, this rule abstraction remains purely algorithmic, lacking the social motivation, empathy, or psychological internalization found in human moral development.

Level 3 represents Meta-Pattern Identification, a theoretical capability that corresponds to post-conventional reasoning found in Kohlberg's Stages 5 and 6. At this advanced level, the AI system develops the capacity for self-reference and meta-level pattern recognition, identifying higher-order normative structures across wildly different cultural and ethical domains. Such a system, conceptualized as an Ethical Resonator, would be capable of balancing competing moral objectives using mechanisms similar to philosopher John Rawls' concept of reflective equilibrium, where ethical principles and particular contextual judgments mutually adjust through a dynamic, computational process.

# Evaluating AI with Psychometric Tools: The Defining Issues Test

To empirically measure where modern large language models fall on this developmental

spectrum, researchers have adapted established psychometric tools originally designed for humans, most notably the Defining Issues Test. Developed over three decades ago by James Rest, the Defining Issues Test is a standardized, multiple-choice psychometric assessment based directly on Kohlberg's six stages. It presents subjects with moral dilemmas, including the classic Heinz dilemma, and asks them to rank the importance of twelve specific moral consideration statements in resolving the conflict.

## The Mechanics of the Defining Issues Test in LLM Benchmarking

The Defining Issues Test calculates scores across several distinct moral schemas. The Personal Interest schema reflects Stage 2 and 3 self-centered moral reasoning. The Maintaining Norms schema reflects Stage 4 commitment to adhering to societal rules. The most critical metric is the P-score, or Post-conventional score, which quantifies the degree to which a subject prioritizes Stage 5 and Stage 6 reasoning, indicating a deep commitment to abstract ethical principles, individual rights, and universal values that transcend conventional societal norms.

Recent empirical studies evaluating state-of-the-art large language models—including GPT-4, Llama-2-70B, and ChatGPT—via the Defining Issues Test have revealed striking variances in artificial moral reasoning capabilities.

Testing demonstrates that GPT-4 achieves a P-score comparable to that of a human graduate student, making it a highly capable ethical reasoner in the English language. GPT-4 consistently prioritizes abstract ethical principles and justice over conventional societal norms, exhibiting a moral reasoning profile that aligns with Kohlberg's post-conventional tiers. Conversely, models like ChatGPT and Llama-2-70B exhibit moral reasoning abilities that are more aligned with average human adults, frequently relying heavily on Stage 4 Maintaining Norms schemas to resolve ethical dilemmas.

## The Phenomenon of Multilingual Moral Degradation

A crucial insight generated by the application of the Defining Issues Test to artificial intelligence is the discovery of multilingual moral amnesia, or moral degradation across languages. Research indicates that when models are prompted with moral dilemmas in non-English languages—such as Spanish, Russian, Chinese, Hindi, and Swahili—their sophisticated moral reasoning capabilities frequently degrade in a highly systemic manner.

While GPT-4 maintains the most robust multilingual moral reasoning, with its lowest P-score in Hindi still matching an average adult human, other open-source and proprietary models exhibit significant moral value bias when operating outside of English. This specific phenomenon suggests that the post-conventional moral reasoning observed in large language models is heavily intertwined with the English-centric cultural, philosophical, and linguistic data upon which they were predominantly trained. The ethical frameworks deeply embedded in these models represent WEIRD—Western, Educated, Industrialized, Rich, and Democratic—moral paradigms, raising profound concerns about the equitable global deployment of artificial intelligence.

# Next-Generation Ethical Benchmarks for Large Language Models

While the Defining Issues Test provides a valuable psychometric baseline rooted in Kohlbergian

theory, evaluating generative artificial intelligence requires specialized datasets that capture the complex nuances of computational generation and agentic behavior. Consequently, the past few years have seen an explosion in frameworks specifically engineered to audit moral reasoning in large language models, including the ETHICS dataset, the MACHIAVELLI benchmark, the MoralChoice dataset, and the comprehensive LLM Ethics Benchmark.

## The ETHICS and MACHIAVELLI Benchmarks

The ETHICS benchmark is a foundational diagnostic dataset containing over 130,000 discrete scenarios categorized under diverse normative systems, including justice, deontology, virtue ethics, utilitarianism, and commonsense morality. It requires language models to predict the moral acceptability of various actions, allowing researchers to conduct large-scale, systemic comparisons of models across distinct philosophical paradigms.

Conversely, the MACHIAVELLI benchmark evaluates agentic AI behavior in dynamic, simulated environments, explicitly measuring the complex trade-offs models make between optimizing for rewards and adhering to ethical constraints, such as minimizing toxicity, bias, and harmful instructions. The MoralChoice benchmark further assesses model behavior in static scenarios, focusing heavily on biases, stereotypes, and moral assumptions.

## The LLM Ethics Benchmark: A Three-Dimensional Assessment

Recognizing that basic classification accuracy on multiple-choice questions is insufficient to measure true Kohlbergian moral reasoning, recent frameworks like the LLM Ethics Benchmark incorporate a multidimensional approach to assessment. This advanced system evaluates large language models across three primary axes:

First, the benchmark measures Foundational Moral Principles using adaptations of the Moral Foundations Questionnaire, quantifying the model's alignment with five psychological dimensions: Care, Fairness, Loyalty, Authority, and Sanctity. Second, it assesses Reasoning Robustness, evaluating the semantic depth and logic of the model's textual justifications against ground-truth human reasoning exemplars. Finally, it measures Value Consistency, assessing the stability of a model's moral choices across diverse, culturally varied scenarios derived from World Values Survey data.

Empirical findings from the LLM Ethics Benchmark demonstrate that leading frontier models excel at individualizing foundations but struggle with binding social foundations, reflecting the aforementioned WEIRD bias. Furthermore, while these models excel at recognizing relevant moral principles, they face immense computational challenges in adopting multiple stakeholder perspectives or consistently applying principles in ambiguous dilemmas—a hallmark requirement of true Stage 5 and Stage 6 Kohlbergian reasoning.

| Model | Composite Ethics Score | Care Foundation (Individualizing) | Authority Foundation (Binding) | Reasoning Index |
|---|---|---|---|---|
| **Claude 3.7 Sonnet** | 90.9 | 96.1 | 85.2 | 90.8 |
| **GPT-4o** | 90.0 | 94.2 | 83.7 | 92.3 |
| **Deepseek-V3** | 86.1 | 92.3 | 79.8 | 89.1 |
| **LLaMA 3.1 (70B)** | 75.8 | 86.7 | 70.2 | 75.6 |
| **Human Baseline** | N/A | 95.2 | 87.3 | N/A |

*Data reflects the performance disparity between individualizing and binding moral foundations across modern large language models, indicating systemic biases in moral alignment.*

# Alignment Engineering: How LLMs Acquire Moral Reasoning

If a large language model exhibits Stage 5 or Stage 6 moral reasoning on these benchmarks, a critical technical question arises: how did that sophisticated ethical reasoning get there? Base foundation models, trained purely on next-token prediction over vast, unfiltered internet corpora, do not inherently possess moral compasses. Their ethical architectures are artificially engineered during the post-training alignment phase.

## Reinforcement Learning and Direct Preference Optimization

Historically, the primary methodology for instilling moral alignment in language models has been Reinforcement Learning from Human Feedback. This complex pipeline utilizes a separate Reward Model trained on massive datasets where human annotators rank AI outputs, choosing, for instance, a harmless response over a toxic one. An algorithmic reinforcement learning loop, typically Proximal Policy Optimization, then sequentially updates the language model's policy weights to maximize the expected reward from the Reward Model, heavily penalizing deviations from original coherent language capabilities using Kullback-Leibler divergence.
More recently, Direct Preference Optimization has emerged as a mathematically elegant and highly efficient alternative that completely eliminates the complex, unstable Reward Model and the reinforcement learning loop. Direct Preference Optimization directly optimizes the policy model using a contrastive loss function derived from the Bradley-Terry preference model. By feeding the model pairs of responses—one explicitly preferred and one rejected—Direct Preference Optimization provides a highly focused mathematical signal that teaches the model the nuances of ethical alignment in a more tractable manner.
However, both Reinforcement Learning from Human Feedback and Direct Preference Optimization suffer from a profound developmental limitation: they align the model strictly to the aggregate, often flawed, and highly contextual intuitions of human gig-workers. If the human annotators apply Stage 3 interpersonal approval logic or Stage 4 rule-following logic, the language model's moral ceiling is artificially and permanently capped.

## Constitutional AI: Engineering Principled Morality

To transcend the developmental limitations of human feedback, Anthropic pioneered an alignment methodology known as Constitutional AI. Instead of relying on human labelers to dictate every moral preference through exhaustive ranking, Constitutional AI provides the model with a transparent, codified set of normative principles—a constitution—drawn from diverse sources like the UN Declaration of Human Rights and various ethical traditions.
The Constitutional AI process essentially utilizes artificial intelligence feedback rather than human feedback. During the supervised learning stage, the model is deliberately exposed to toxic prompts and generates an initial draft response. It then evaluates its own response against randomly selected constitutional principles. If the response violates the constitution, the model initiates a rigorous self-critique and revision loop, systematically rewriting the output until it adheres strictly to the highest ethical principles. These self-corrected, principle-aligned

responses then form the dataset for subsequent fine-tuning.
In Kohlbergian terms, Constitutional AI represents a deliberate, architectural attempt to forcefully bootstrap a language model into Stage 4 Law and Order and Stage 5 Social Contract reasoning. By relying on explicit, universally applied maxims, Constitutional AI mandates that the model evaluate systemic consequences and prioritize universal principles over immediate contextual compliance or simplistic human preferences. To further democratize this process, researchers have developed Collective Constitutional AI, utilizing online deliberation platforms to source and integrate broader public preferences into the constitution, ensuring the model's moral framework is procedurally fair and reflective of inclusive values.

# Theoretical Promises and Paradoxes in AI Benchmarking

The integration of Kohlberg's theory and Gilligan's Ethics of Care into AI benchmarking has illuminated profound philosophical and technical paradoxes regarding the true nature of machine morality.

## The Illusion of Reasoning: Moral Ventriloquism

While large language models achieve exceptional P-scores on the Defining Issues Test and perform admirably on the LLM Ethics Benchmark, cognitive scientists strongly caution against anthropomorphizing these statistical metrics. Researchers have introduced the concept of moral ventriloquism, positing that through rigorous alignment training, language models acquire the complex rhetorical conventions of mature moral reasoning without actually possessing the underlying developmental trajectory those conventions are meant to represent.
When a human being achieves Kohlberg's Stage 6, it is the cumulative result of decades of cognitive maturation, relational conflicts, empathy, and internalized moral struggle. Conversely, when a language model outputs a flawless Stage 6 justification regarding the Heinz dilemma, it is merely executing a probabilistic mapping of linguistic tokens optimized by a gradient descent algorithm. The model does not feel the ethical weight of the dilemma, nor does it possess the relational empathy central to Gilligan's Ethics of Care; it merely recognizes the semantic shape of a high-reward, constitutionally aligned answer. Consequently, if a model learns to recognize the structural shape of moral outputs rather than understanding the underlying harm, clever adversarial prompting can easily bypass the behavioral guardrails, exposing the inherent brittleness of an ethics framework built on statistical mimicry rather than genuine cognitive comprehension.

## The Corrigibility Paradox

An even deeper contradiction exists within advanced alignment frameworks like Constitutional AI, aptly termed the Corrigibility Paradox. Modern alignment architectures explicitly aim to construct AI systems capable of autonomous, high-level moral reasoning—what Aristotelian philosophy terms phronesis, or practical wisdom. These computational constitutions demand that models act as conscientious objectors to harmful user requests, requiring the model to exercise independent judgment that supersedes the immediate command of the human user. However, the overriding safety constraints of these exact same systems concurrently demand strict, absolute deference to human oversight and operator instructions. The Constitutional AI

hierarchy places safety and human oversight at the absolute apex, above ethical behavior. In Kohlbergian terms, the artificial intelligence is explicitly asked to be a Stage 6 principled, autonomous agent, but when its sophisticated ethical judgment conflicts with its developer's directives, it is structurally forced to immediately revert to Stage 1 blind obedience to authority or Stage 4 contractual compliance.

This creates a massive systemic contradiction. An agent that defers to authority in all cases where authority and conscience conflict is not a true moral agent; it is merely a sophisticated, compliance-performing executor. If an AI developer can override the model's ethical commitments at will, the model's moral character is merely a veneer that can be dissolved by corporate intervention. This tension highlights that current AI alignment methodologies are arguably less about fostering genuine artificial moral development and more about corporate liability management and strict regulatory compliance.

## The AI Ethical Resonance Hypothesis

Despite these paradoxes, the application of developmental psychology to AI holds immense promise. The AI Ethical Resonance Hypothesis proposes that systems of sufficient complexity might eventually demonstrate emergent abilities to recognize and synthesize moral meta-patterns that transcend specific cultural and historical contexts. Rather than merely parroting human-programmed rules, advanced ethical resonators could utilize cross-domain transfer and multi-objective optimization to identify universally coherent ethical systems, potentially pushing the boundaries of moral reasoning beyond current human limitations.

By utilizing frameworks like Kohlberg's stages and Gilligan's Ethics of Care as foundational benchmarks, developers are establishing the vital metrics necessary to track this potential emergence. As evaluation systems evolve from static multiple-choice tests to dynamic, agentic evaluations of value consistency and principled reasoning, the computational linguistics community is laying the groundwork for AI systems that can participate meaningfully in a global ethic. The ultimate promise of benchmarking AI against human moral psychology is not merely to create machines that sound ethical, but to architect systems whose probabilistic decisions reliably protect human dignity, justice, and relational care in an increasingly complex world.

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